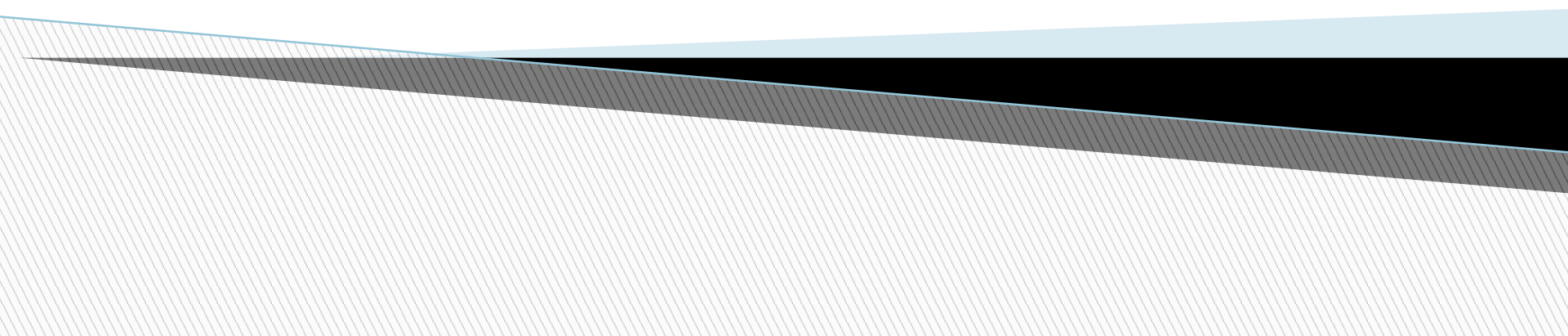


Project Cost Management



The Importance of Project Cost Management

- ? IT projects have a poor track record for meeting budget goals
- ? A cost **overrun** is the additional percentage or dollar amount by which actual costs exceed estimates
- ? A 2011 Harvard Business Review study reported an average cost overrun of 27 percent. The most important finding was the discovery of a large number of gigantic overages or “black swans”

What is Cost and Project Cost Management?

- ? **Cost** is a resource sacrificed or foregone to achieve a specific objective or something given up in exchange
- ? Costs are usually measured in monetary units like dollars
- ? **Project cost management** includes the processes required to ensure that the project is completed within an approved budget

Project Cost Management Processes

- ? **Planning cost management** :determining the policies, procedures, and documentation that will be used for planning, executing, and controlling project cost.
- ? **Estimating costs**: developing an approximation or estimate of the costs of the resources needed to complete a project
- ? **Determining the budget**: allocating the overall cost estimate to individual work items to establish a baseline for measuring performance
- ? **Controlling costs**: controlling changes to the project budget

Figure 7-1. Project Cost Management Summary

Planning

Process: **Plan cost management**

Outputs: Cost management plan

Process: **Estimate costs**

Outputs: Activity cost estimates, basis of estimates, project documents updates

Process: **Determine budget**

Outputs: Cost baseline, project funding requirements, project documents updates

Monitoring and Controlling

Process: **Control costs**

Outputs: Work performance information, cost forecasts, change requests, project management plan updates, project documents updates, organizational process assets updates

Project Start

Project Finish

Types of Costs and Benefits

- ? **Tangible costs or benefits** are those costs or benefits that an organization can easily measure in dollars
- ? **Intangible costs or benefits** are costs or benefits that are difficult to measure in monetary terms
- ? **Direct costs** are costs that can be directly related to producing the products and services of the project
- ? **Indirect costs** are costs that are not directly related to the products or services of the project, but are indirectly related to performing the project
- ? **Sunk cost** is money that has been spent in the past; when deciding what projects to invest in or continue, you should *not* include sunk costs

More Basic Principles of Cost Management

- ? **Learning curve theory** states that when many items are produced repetitively, the unit cost of those items decreases in a regular pattern as more units are produced
- ? **Reserves** are dollars included in a cost estimate to mitigate cost risk by allowing for future situations that are difficult to predict
 - **Contingency reserves** allow for future situations that may be partially planned for (sometimes called **known unknowns**) and are included in the project cost baseline
 - **Management reserves** allow for future situations that are unpredictable (sometimes called **unknown unknowns**)

Planning Cost Management

- ? The project team uses expert judgment, analytical techniques, and meetings to develop the cost management plan
- ? A cost management plan includes:
 - Level of accuracy and units of measure
 - Organizational procedure links
 - Control thresholds
 - Rules of performance measurement
 - Reporting formats
 - Process descriptions

Estimating Costs

- ? Project managers must take cost estimates seriously if they want to complete projects within budget constraints
- ? It's important to know the types of cost estimates, how to prepare cost estimates, and typical problems associated with IT cost estimates

Table 7-1. Types of Cost Estimates

TYPE OF ESTIMATE	WHEN DONE	WHY DON'T	HOW ACCURATE
Rough order of magnitude (ROM)	Very early in the project life cycle.	Provides estimate of cost for selection decisions	-50% to +100%
Budgetary	During project planning or when scope is clearer	Puts dollars in the budget plans	-10% to +25%
Definitive	Once project design and scope are well defined (execution planning phase)	Provides details for purchases, estimates actual costs	-5% to +10%

Cost Estimation Tools and Techniques

- ? Basic tools and techniques for cost estimates:
 - **Analogous or top-down estimates:** use the actual cost of a previous, similar project as the basis for estimating the cost of the current project
 - **Bottom-up estimates:** involve estimating individual work items or activities and summing them to get a project total
 - **Parametric modeling** uses project characteristics (parameters) in a mathematical model to estimate project costs

Typical Problems with IT Cost Estimates

- ? Estimates are done too quickly
- ? People lack estimating experience
- ? Human beings are biased toward underestimation
- ? Management desires accuracy

Sample Cost Estimate

- ? See the text for a detailed example of creating a cost estimate for the Surveyor Pro project described in the opening case
- ? Before creating an estimate, know what it will be used for, gather as much information as possible, and clarify the ground rules and assumptions for the estimate
- ? If possible, estimate costs by major WBS categories
- ? Create a cost model to make it easy to make changes to and document the estimate

Figure 7-2. Surveyor Pro Project Cost Estimate

Surveyor Pro Project Cost Estimate October 5

	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	WBS Level 2 Totals	% of Total
WBS Items					
1. Project Management				\$306,300	20%
Project manager	960	\$100	\$96,000		
Project team members	1920	\$75	\$144,000		
Contractors (10% of software development and testing)			\$66,300		
2. Hardware				\$76,000	5%
2.1 Handheld devices	100	\$600	\$60,000		
2.2 Servers	4	\$4,000	\$16,000		
3. Software				\$614,000	40%
3.1 Licensed software	100	\$200	\$20,000		
3.2 Software development*			\$594,000		
4. Testing (10% of total hardware and software costs)			\$69,000	\$69,000	5%
5. Training and Support			Empty cell	\$202,400	13%
Trainee cost	100	\$500	\$50,000		
Travel cost	12	\$700	\$8,400		
Project team members	1920	\$75	\$144,000		
6. Reserves (20% of total estimate)			\$253,540	\$253,540	17%
Total project cost estimate				\$1,521,240	

*See software development estimate

Figure 7-3. Surveyor Pro Software Development Estimate

1. Labor Estimate	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	Calculations
Contractor labor estimate	3000	\$150	\$450,000	3000*150
Project team member estimate	1920	\$75	\$144,000	1920*75
Total labor estimate			\$594,000	Sum above two values
Empty cell				
2. Function point estimate	Quantity	Conversion Factor	Function Points	Calculations
External inputs	10	4	40	10*4
External interface files	3	7	21	3*7
External outputs	4	5	20	4*5
External queries	6	4	24	6*4
Logical internal tables	7	10	70	7*10
Total function points			175	Sum above function point values
Java 2 language equivalency value			46	Assumed value from reference
Source lines of code (SLOC) estimate			8,050	175*46
ProductivityxKSLOC^Penalty (in months)			29.28	3.13*8.05^1.072 (see reference)
Total labor hours (27 hours/function point)*			4,725	27*175
Cost/labor hour (\$120/hour)			\$120	Assumed value from budget expert
Total function point estimate			\$567.000	4,725*120

*Based on historical data

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Determining the Budget

- ? Cost budgeting involves allocating the project cost estimate to individual work items over time
- ? The WBS is a required input to the cost budgeting process since it defines the work items
- ? Important goal is to produce a **cost baseline**
 - a time-phased budget that project managers use to measure and monitor cost performance

Controlling Costs

- ? Project cost control includes
 - Monitoring cost performance
 - Ensuring that only appropriate project changes are included in a revised cost baseline
 - Informing project stakeholders of authorized changes to the project that will affect costs
- ? Many organizations around the globe have problems with cost control

Project Portfolio Management

- ? Many organizations collect and control an entire suite of projects or investments as one set of interrelated activities in a portfolio
- ? Five levels for project portfolio management
 1. Put all your projects in one database
 2. Prioritize the projects in your database
 3. Divide your projects into two or three budgets based on type of investment
 4. Automate the repository
 5. Apply modern portfolio theory, including risk-return tools that map project risk on a curve

Summary

- ? Project cost management is a traditionally weak area of IT projects, and project managers must work to improve their ability to deliver projects within approved budgets
- ? Main processes include
 - Plan cost management
 - Estimate costs
 - Determine the budget
 - Control costs